Paper 4B/4

THE IRRADIATION BEHAVIOUR OF PLUTONIUM-BEARING CERAMIC FUEL PINS

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1. Introduction

Since the Dounreay fast reactor went to full power, it has been extensively used as an irradiation test facility to examine possible designs of fuel element for the prototype fast reactor. This paper summarizes the fuel pin irradiation programme, and indicates the principal results obtained by the autumn of 1965.

Irradiation testing has concentrated on co-precipitated mixed oxide (85% UO_2, 15% PuO_2). A modified version of type 316 stainless steel (nominal composition 17% Cr, 13·5% Ni, 2·5% Mo), usually of low carbon content (<0·03%) has been used for cladding for all but one pin for which Nimonic 80A was used. Pins were mainly of 5·08 mm internal diameter, 5·84 mm external diameter, but with a very few tests of 6·98–8·25 mm diameter. Both vibro-compacted and annular-pelleted fuel have been used, at loading densities of about 80% of the theoretical maximum. The remaining space was deliberately left to accommodate solid fission products, and provide a path for released gaseous fission products to a plenum at the top of the pin. The ratio of fuel to plenum length has been calculated from considerations of the mechanical properties of the cladding: in particular it has been arranged that the cladding would neither yield, nor creep by more than 0·1%, during its service life under the joint action of the stresses set up by (1) the gaseous internal pressure and (2) the high heat flux through the can. Data on unirradiated cladding were used for these calculations.

The maximum temperature of the cladding has been in the range 550–650°C. Ratings in the range 100–250 W/g have been used, with emphasis on the upper end. This has been achieved in DFR by using fuel containing

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enriched uranium as well as 15% plutonium. Pins have been designed to run to 5% or 10% mean burnup of heavy atoms* and results are quoted here up to 7·6% mean burnup.

TABLE 1. FABRICATION PARAMETERS AND IRRADIATION

Pin no.	Fabrication parameters					
	Pin length (cm)	Pin diameter o.d./i.d. (cm)	Cladding	Void/ fuel ratio	Fuel length (cm)	Pack density (% T.D.)
V.166	17·36	0·584/0·508	M316L, C.W.	0·90	9·30	85·3
V.024	17·36	0·584/0·508	M316L, S.T.	1·56	6·48	79·9
V.021	17·38	0·584/0·508	M316L, S.T.	0·93	9·02	81·7
V.162 V.162R	71·11	0·584/0·508	M316L, C.W.	0·52	50·55	84·7
V.035 V.035R	71·09	0·584/0·508	M316L, S.T.	0·58	50·30	82·3
V.037	71·09	0·584/0·508	M316L, S.T.	0·58	49·99	82·5
V.039 V.039R	71·15	0·584/0·508	M316L, C.W.	0·55	50·80	82·6
V.173	71·11	0·584/0·508	M316L, S.T.	0·95	41·65	82·1
A.151 A.151R	71·09	0·584/0·508	M316L, C.W.	0·60	51·05	72·4
A.153 A.153R	71·10	0·584/0·508	M316L, C.W.	0·58	51·02	72·9
V.177 V.177R	63·27	0·584/0·508	Nim. 80A, F.H.T.	0·34	52·55	83·2
V.233	76·70	0·584/0·508	M316L, C.W.	Vented	51·05	83·3
V.259	58·24	0·584/0·508	M316L, C.W.	0·44	46·23	79·3
V.258	71·15	0·825/0·698	M316L, C.W.	0·73	46·42	81·6

Notes:

(1) Prefixes V, A, in pin reference No. indicate vibro-compacted and annular pellet fuel, respectively. The pellet i.d. was 0·090.

(2) 'R' after pin no. indicates re-encapsulated pin.

(3) Cladding condition indicated by abbreviations:

S.T. = solution treated, i.e. 20 min at 1050°C, as last stage of fabrication.

C.W. = 20% cold worked, condition in which tube is received from manufacturer.

F.H.T. = fully heat treated, i.e. S.T. at 1080°C for 30 min, age 16 hours at 700°C.

The examinations of fourteen oxide pins irradiated in DFR to mean burnups in excess of 5% are summarized in this paper. Four of these were examined non-destructively and returned to the reactor for further irradia-

* It is perhaps helpful to note that 1% burnup of heavy atoms is equivalent to about 8200 MWd/t of oxide, and 8800 MWd/t of carbide.

tion. Three of the oxide pins failed by longitudinal cracking of the can; the remainder, with one possible exception, appeared to be intact. Failure mechanisms are discussed in the light of the evidence obtained.

CONDITIONS FOR PINS IRRADIATED TO HIGH BURNUP IN DFR

Irradiation conditions					Condition on discharge	Post-discharge history
Max. rating (W/g)	$\int kd\theta$ (max.) initial (W/cm)	Burnup % heavy atoms		Clad. temp. (predicted) (°C)		
		Max.	Mean			
182	26	5·8	5·8	665	Intact	Destructively examined
170	24	7·1	7·1	700	Failed	Destructively examined
210	29	8·1	8·1	655	Failed	Destructively examined
230	36	5·1	4·3	590	Intact	N.D.T., re-encapsulated
222		7·1	6·2	565	Intact	Destructively examined
175	26	5·2	4·5	560	Intact	N.D.T., re-encapsulated
170		6·9	6·1	625	Possible failure	N.D.T., pierced, stored
172	25	6·9	6·1	555	Intact	N.D.T., re-encapsulated
180	26	5·7	5·1	560	Intact	N.D.T., re-encapsulated
177		7·5	6·6	625	Intact	N.D.T., pierced, stored
182	27	6·7	6·1	595	Intact	N.D.T., re-encapsulated
200	14	5·8	5·1	570	Intact	N.D.T., re-encapsulated
191	14	7·7	6·8	630	Intact	N.D.T., re-encapsulated
224	17	7·0	5·8	560	Intact	N.D.T., re-encapsulated
216	17	9·1	7·6	625	Failed	Destructively examined
165	24	2·9	2·6	650	Intact	N.D.T., re-encapsulated
163	24	5·8	5·3	645	Intact	N.D.T., re-encapsulated
227	33	6·0	5·0	570	Intact	N.D.T., re-encapsulated
224	33	6·0	5·2	435	Intact	N.D.T., re-encapsulated
224	66	6·0	5·1	650	Intact	N.D.T., pierced, re-sealed, re-encapsulated

(4) Void/fuel ratio given by ratio total volume of voidage relative to fuel alone.
(5) Theoretical density of mixed oxide fuel assumed to be 10·92 g/cm^3.
(6) Ratings and clad temperatures apply to conditions prevailing at terminal burnup.
(7) Maximum clad temperatures were raised in some cases by adjustment of flow rate through vehicle during re-encapsulation.

2. Experimental Details

Pin and Vehicle Design

Details of the pin dimensions, cladding materials, fuel characteristics, and fuel loadings, together with data on irradiation conditions to which the pins were subjected, are listed in Table 1.

Manufacture of these pins followed a common route despite variations in dimensions and component materials. After welding the bottom end cap to the cladding tube, insertion follows in order of a steel pellet—filling the cavity in the bottom end plug—a sintered natural UO_2 insulator pellet, the

TABLE 2. SUMMARY OF MENSURATION DATA ON FUEL PINS IRRADIATED IN THE DFR CORE

Pin no.	Max. burnup (%)	Ovality (cm)	Max. dia. change (%)	Change at plenum (%)	Pin extension (%)
V.019	0·21	—	−0·16	Nil	—
V.020	1·78	—	−0·32	Nil	—
V.025	2·67	—	−0·10	Nil	—
V.166	5·8	—	0·65	0·1	—
V.024	7·1	—	1·0*	0·8	0·6
V.021	8·1	—	1·7†	0·8	0·34
V.036	0·2	0·0015	0·44	—	—
V.050	0·9	—	−0·44	—	—
V.285	2·2	0·0033	0·5	0·1	0
V.162	5·1	—	0·3	0–0·15	0·007
V.162R	7·1	0·0025	0·63	0–0·2	0·042
V.035	5·2	0·0023	0·35‡	0·1	0·05
V.035R	6·9	0·0023	0·7‡	Nil	0·025
V.039	5·7	—	0·42	0·05	0·09
V.039R	7·5	0·0015	0·76	0·05	0·12
V.173	6·7	0·0018	0·6	Nil	0·07
V.037	6·9	—	0·52	Nil	0·18
V.233	6·0	0·0010	0·56	—	0·014
V.258	6·0	0·0015	0·85	0·1	0·04
V.259	6·0	0·0015	0·65	—	0·05
V.177	2·9	0·0008	0·25	—	−0·05
V.177R	5·8	0·008	0·25	—	—
A.031	1·0	0·008	0·36	—	—
A.041	1·8	+0·008	−0·14	—	—
A.242	2·2	0·0018	0·43	Nil	—
A.213	2·2	0·008	0·35	Nil	—
A.151	5·8	0·0013	0·14	Nil	0·035
A.151R	7·7	0·0038	0·5	Nil	0·046
A.153	7·0	—	1·3	—	−0·025
A.153R	9·1	0·0100	1·7*	—	0·02

* Remote from cracks.
† Between cracks.
‡ At base.

enriched fuel, a second insulator pellet, and a stainless steel collet bearing a central strut, designed to prevent movement of fuel up the plenum should conditions favour ratcheting. An expansion space is normally left between the top of the strut and the end cap, and the width of this gap serves as a sensitive indicator of fuel movement. The top cap is welded on in an atmosphere of helium (purity 99%) and the cladding is then heat-treated as necessary.

Two basic types of irradiation vehicle for testing pins have been used in DFR. One type which transfers the heat from the pin radially outwards is exemplified by the rig in which a pin 17·8 cm long is immersed in a static pot of sodium. In the second type, the heat is removed by reactor NaK flowing past the pin. Pins 71·1 cm in length have been tested in this way. The pin

TABLE 3. FISSION GAS RELEASE FROM MIXED OXIDE PINS IRRADIATED IN DFR CORE

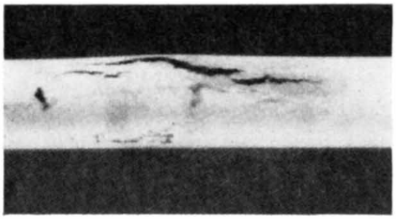
Pin no.	Type	Mean burnup (%)	Max. final rating (W/g)	Gas release			Gas retained by fuel (cm^3/g)
				Xe (%)	Kr (%)	Total (%)	
V.019	17·8 cm, 5·08 mm i.d.	0·21	173	36	34	36	0·028
V.020	17·8 cm, 5·08 mm i.d.	1·78	177	50	56	51	0·183
V.026	17·8 cm, 5·08 mm i.d.	2·63	192	59	66	60	0·220
V.166	17·8 cm, 5·08 mm i.d.	5·8	192	79	83	80	0·245
V.024*	17·8 cm, 5·08 mm i.d.	7·1	170				
V.021*	17·8 cm, 5·08 mm i.d.	8·1	210				
V.036	71·1 cm, 5·08 mm i.d.	0·17	173	21	20	21	0·028
V.162R	71·1 cm, 5·08 mm i.d.	6·2	230	70	74	71	0·378
V.035R	71·1 cm, 5·08 mm i.d.	6·1	170			76–85	0·19–0·31
V.039R	71·1 cm, 5·08 mm i.d.	6·6	180	80	85	81	0·270
V.050	58·4 cm, 5·08 mm i.d.	0·78	175	43	47	44	0·090
V.259	58·4 cm, 5·08 mm i.d.	5·2	224	78	80	78	0·243
V.257	71·1 cm, 6·98 mm i.d.	1·53	220	81	83	82	0·060
V.258	71·1 cm, 6·98 mm i.d.	5·1	220	91	96	92	0·092
A.031	71·1 cm, 6·98 mm i.d.	0·83	237	5	5	5	0·166
A.041	71·1 cm, 6·98 mm i.d.	1·6	176	33	28	33	0·218
A.151R	71·1 cm, 6·98 mm i.d.	6·8	191	44	47	44	0·807

* 36% and 45% of gases produced in V.024 and V.021, respectively, were collected on piercing the sodium filled capsule and later the pin itself: losses are indeterminable.

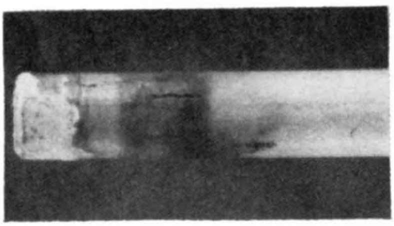
temperature varies from 230°C (reactor coolant inlet temperature) at the top to a maximum figure at the bottom determined by the flow of NaK through a calibrated orifice at the top of the rig. Radial heat losses from the pin to the bulk reactor coolant are minimized by the interposition of an argon filled gap. A wire grid cage plus a spider plate at its top end centralize the pin in the flow tube.

Standard post-irradiation examination techniques have been used. One valuable innovation has been the use of re-encapsulation: after irradiation, a pin has been examined non-destructively (measurement, gamma scanning, X-radiography) and then reloaded in a new vehicle for further irradiation. Occasionally the fission product gases have been taken from a pin, the pin resealed, and then re-encapsulated for further irradiation. Several of the pins described here have thus been examined twice.

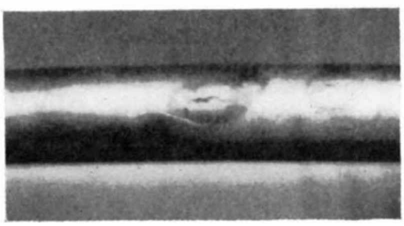
Before sectioning a pin for metallography, it is punctured and its fission product gas content measured. A detailed examination was made of both fuel and cladding, and relevant results are reported below.



a. Pin V.021, 8·1% burnup middle cracks. ×2½



b. Pin V.021, 8·1% burnup cracks at UO_2 pellet. ×2½.



c. Cracks at centre of pin V.024, 7·1% burnup ×2½.

Fig. 1. Appearance of cracks in pins V.021 and V.024.

3. Results and Discussion

The results of the examination of fourteen pins irradiated to a mean burnup in excess of 5% of heavy atoms are summarized in the following sections and in Tables 2 and 3, Figs. 1 to 12. Of these fourteen pins, eleven were seen to be in excellent condition on attaining their scheduled burnup. Only in the case of three pins, which were deliberately irradiated beyond their nominal endurance limit, was failure observed but because of the exceptional interest of features associated with pin failure, data from these pins are presented in especial detail (see Figs. 1 to 3). Where relevant, data from pins which are not listed in Table 1 have been included in other tables and figures.

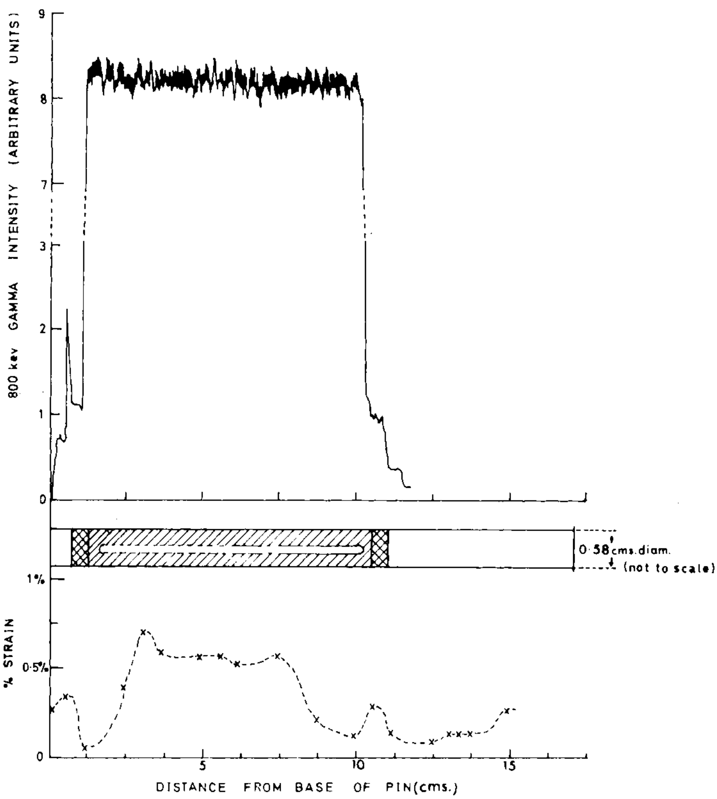


FIG. 2a. Fuel distribution and pin diameter changes in pin V.166.

Non-destructive Examination

Radiography of the irradiation vehicles before breakdown started gave assurance in every case that, fuel pins apart, the various components of the rig had maintained their original conformation and disposition. Intimations were also obtained, from the appearance of swellings, concavities or bends in the fuel surface, of the probable failure of two short fuel pins, V.021 and V.024, and one long pin A.153. Visual inspection of these pins after removal from the vehicle confirmed that they had indeed failed, the cracks in the three pins taking the form:

V.021. Two sets of multiple cracks: (i) at the midpoint of the fuel section a devious crack, about 1 cm long, flanked by two smaller cracks of a similar type associated with a perceptible blister on the clad (Fig. 1a); (ii) at the base

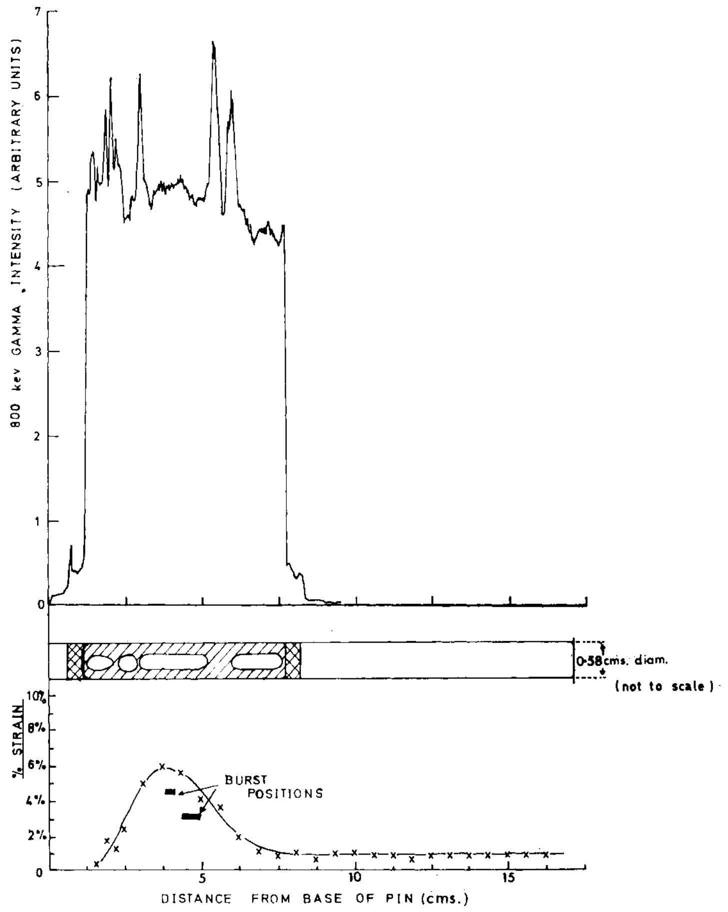


FIG. 2b. Fuel distribution and pin diameter changes in pin V.024.

of the pin, the cladding over the natural UO_2 pellet was blackened (Fig. 1b) but one easily visible crack, bifurcating at its lower end, and four short, straight, narrow cracks running parallel to one another over an arc of about 150° could be distinguished.

V.024. Two small irregular cracks, the longest 5 mm long situated on the apex of a bend coincident with the midpoint of the fuel length (Fig. 1c).

A.153. Three prominent cracks, approximately in line, located at points 5 cm, 25 cm, and 40 cm from the lower (and hotter) end of the pin, the characteristics of each being (see Fig. 3):

(i) a short straight crack or split 1·25 cm long at the base of the pin;
(ii) a longer split, more than 2·5 cm long, plus a shorter narrower crack; displaced 60° round the circumference, at the fuel mid-length;
(iii) an irregular crack or tear, the sides of which were widely separated and appeared to be displaced vertically as well as horizontally.

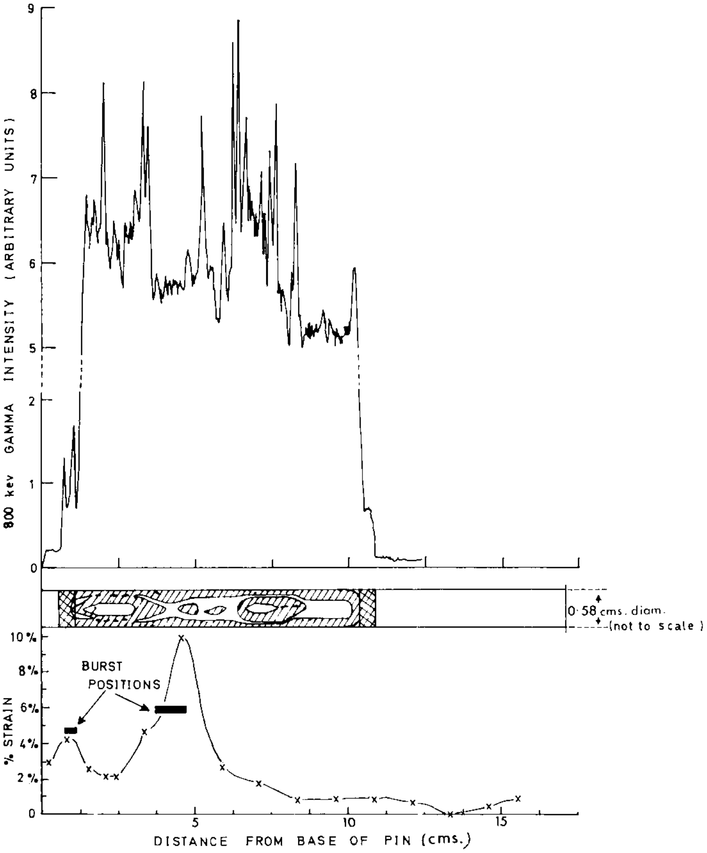


FIG. 2c. Fuel distribution and pin diameter changes in pin V.021

In no other pin was any evidence of cladding damage or interaction with the coolant observed.

Radiographic techniques, applied to the pin alone, gave detailed information on fuel contours and regularity of can/fuel contact; fuel cracking and

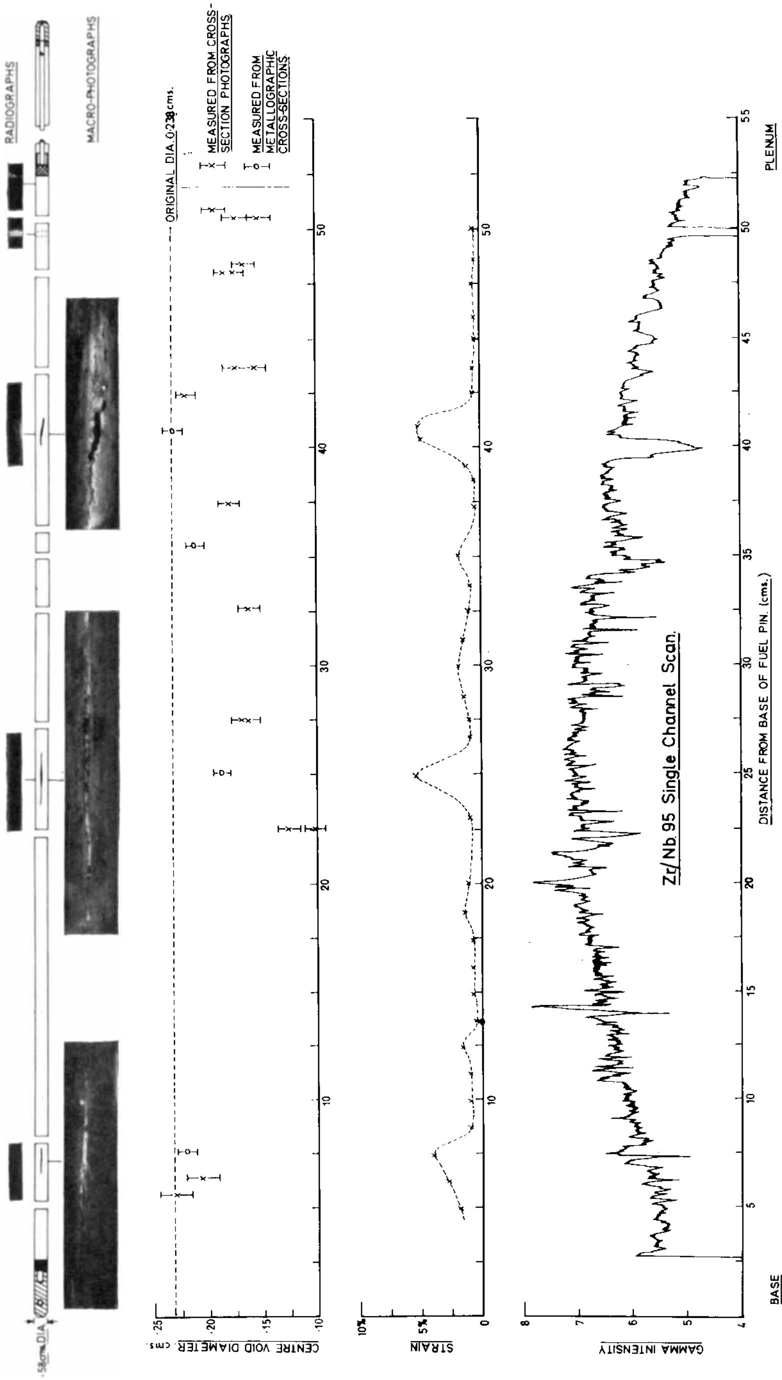


FIG. 3. Post-irradiation examination data, pin A.153 (7·6% mean burnup).

inter-pellet gaps; regions of low density, including the centre void configuration; fuel extension, and pin bowing. Gamma scanning also proved a sensitive means of discriminating discontinuities in the fuel column, and served as a valuable adjunct to radiography in this respect. The major functions of this technique, however, were to provide qualitative information on (a) burnup variation along the pin length and (b) fuel or fission product movements within the pin. Generally, the gamma scan traces for 5·08 mm internal diameter (i.d.) pins had a reasonably smooth, symmetrical quasi-cosine form, simply reflecting the axial flux distribution in the reactor core. Comparison of the scans from the same pin at different stages of burnup (Fig. 4a) suggests that the fuel achieves a stable configuration in the early stages of irradiation which is maintained at burnups beyond 6%. In one case (pin V.162) deviations from the expected cosine form, suggestive of localized fuel slumping, which were observed at low burnup were entirely eliminated on further irradiation (Fig. 4b). In two pins, V.177 and V.039, perturbations in the gamma scan trace evident at lower burnup extended over a greater length of pin after higher burnup (the V.039 case is illustrated in Fig. 4c). In no case could irregularities in the gamma scans of intact pins be correlated with inhomogeneity of fuel distribution, radiography showing no intensity variations, and it is possible that the gamma activity variation is associated entirely with fission product movements.

The form of the gamma scans in the three short pins, V.166, V.024, and V.021 (Fig. 2), appears at first sight to represent consecutive stages in the development of a failure condition, the smooth straight trace of the intact pin changing to a peak and gully trace in V.024 and to a trough and plateau conformation in V.021. Radiographic and metallographic examination has confirmed that the regions of high gamma activity correspond with zones of enhanced fuel concentration, but careful evaluation of the fuel distribution in each failed pin suggests a loss of fuel from the top of the pin and its descent to trapping points on either side of the fracture zone. The fuel distribution seen at the end of irradiation, therefore, would not conduce to cladding failure at the points observed, and may rather be a consequence of failure promoted by gas blanketing of the upper parts of the pin.

Evidence of long range longitudinal fission product movement in 5·08 mm i.d. pins has been confined to indications of caesium and iodine concentration at either end of the pin and of traces of caesium in the plenum. There has been no evidence of any blockage of the insulator pellets in the pins. Isotopic analysis of the caesium in the plenum reveals that it is composed predominantly of Cs^{133}, daughter product of Xe^{133} (half life 5·27 days), the daughter products of Xe^{135} and Xe^{137}, half lives 9·2 hours and 3·8 min respectively, being present in comparatively small concentration. Since the yields of these isotopes are about equal, it is concluded that caesium reaching the plenum arrives almost exclusively as its gaseous precursor, and that, on average, the

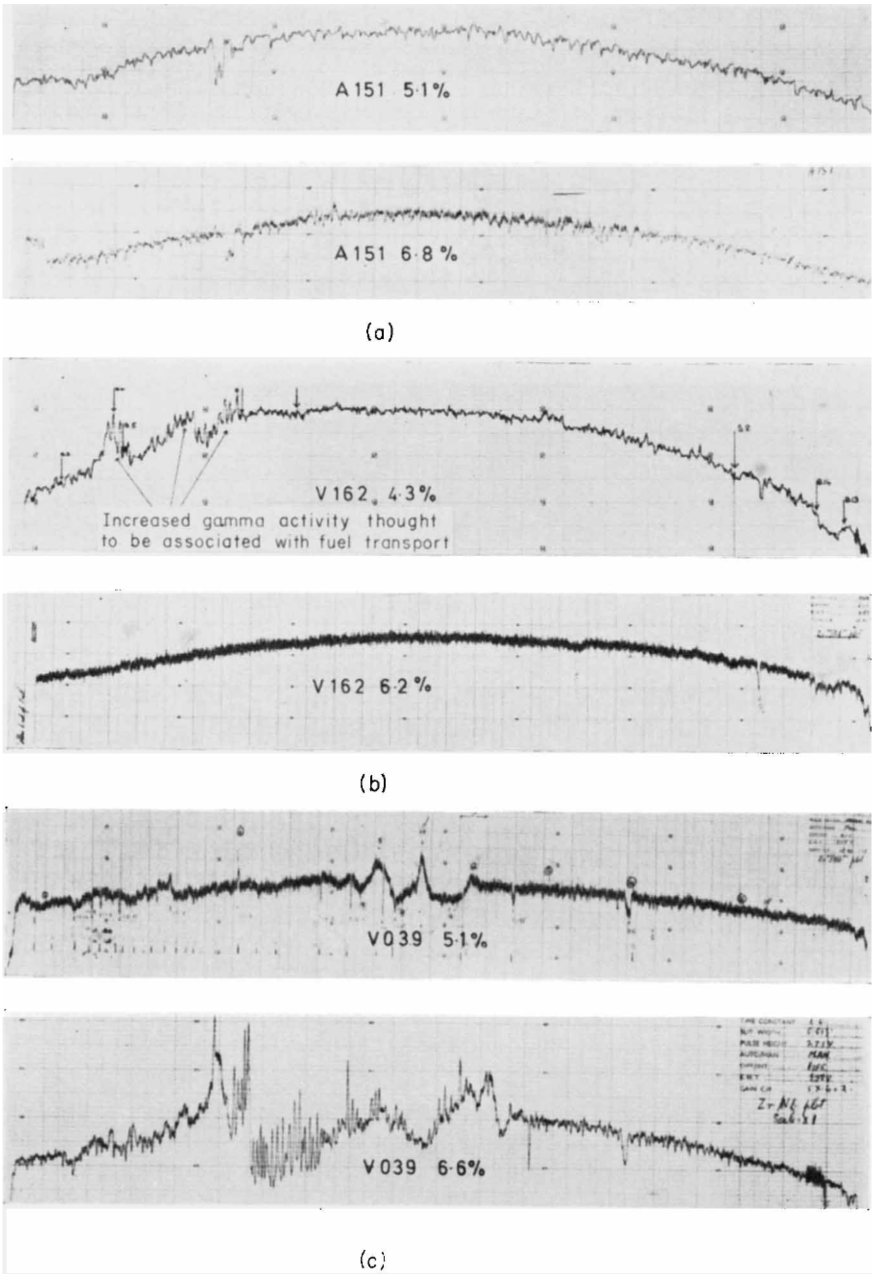


FIG. 4. Gamma scans from fuel pins at successive stages of irradiation.

period elapsing between generation in the fuel and migration into the plenum is about 20 hours. Evidence for fission product migration in 6·98 mm i.d. vibro-compacted oxide pins is much more striking, less volatile fission products such as ruthenium and strontium as well as barium and caesium appearing to move freely, a consequence probably of the higher fuel temperatures.

Pin Measurements

A summary of diameter and length changes is presented in Table 2, and the mean diameter at the high spots in the pins plotted against burnup in Fig. 5. Detailed diameter changes in the short pins and pin A.153 are also included in Figs. 2 and 3.

The uncertainty in the changes in diameter occurring during irradiation is considerably greater than the estimated precision of measurement, 5×10^{-3} mm (0·1%), would suggest. Thus pins measured after low burnup show a scatter in diametral strain of $\pm 0{\cdot}5\%$ which cannot be related to variations in irradiation or material conditions and must be ascribed to cumulative error arising from pin scoring, relaxation of internal stresses, temperature uncertainty during measurement and instrument calibration errors. However, there is less reason to doubt the validity of comparisons made between measurements at successive stages of irradiation, and points from the same pin after increasing periods of irradiation are joined by dotted lines in Fig. 5. It is evident that the increase in diameter in the burnup range 4·5% to 7% is considerable, corresponding to an average of about 0·2% diameter increase per 1% burnup. This represents a strain rate one to two orders of magnitude higher than could be predicted from considerations of stresses in the clad caused by fission gas pressures alone, and suggests either:

(i) an impressive acceleration of creep in stainless steels under irradiation—which is not borne out by complementary studies on this topic; or

(ii) the intervention of an alternative swelling mechanism, i.e. fuel swelling caused by the accumulation of solid and gaseous fission products in the ceramic matrix.

The apparent incubation period of $\sim 3\%$ burnup, in which period fuel swelling could be accommodated by voidage in the fuel, tends to favour the second rather than the first alternative.

Diameter increases in the failed pins are strongly influenced by the width of the cracks. Subtraction of the increment attributable to the crack indicates a strain before fracture of 1·5% to 2% in pin A.153, 4% to 6% at the mid-points of V.021 and V.024, about 1% near the cracks at the base of V.021. The strain profile in the two short pins clearly shows the influence of the temperature gradients along the pin, the maximum strain occurring at the point of maximum temperature. In pin A.153, on the other hand, strain appears to be completely insensitive to clad temperature or burnup, strain maxima occurring at locations where the clad temperature (and corresponding burnup) are ~360°C (8·0% burnup), ~490°C (9·1% burnup) and ~600°C (7·2% burnup).

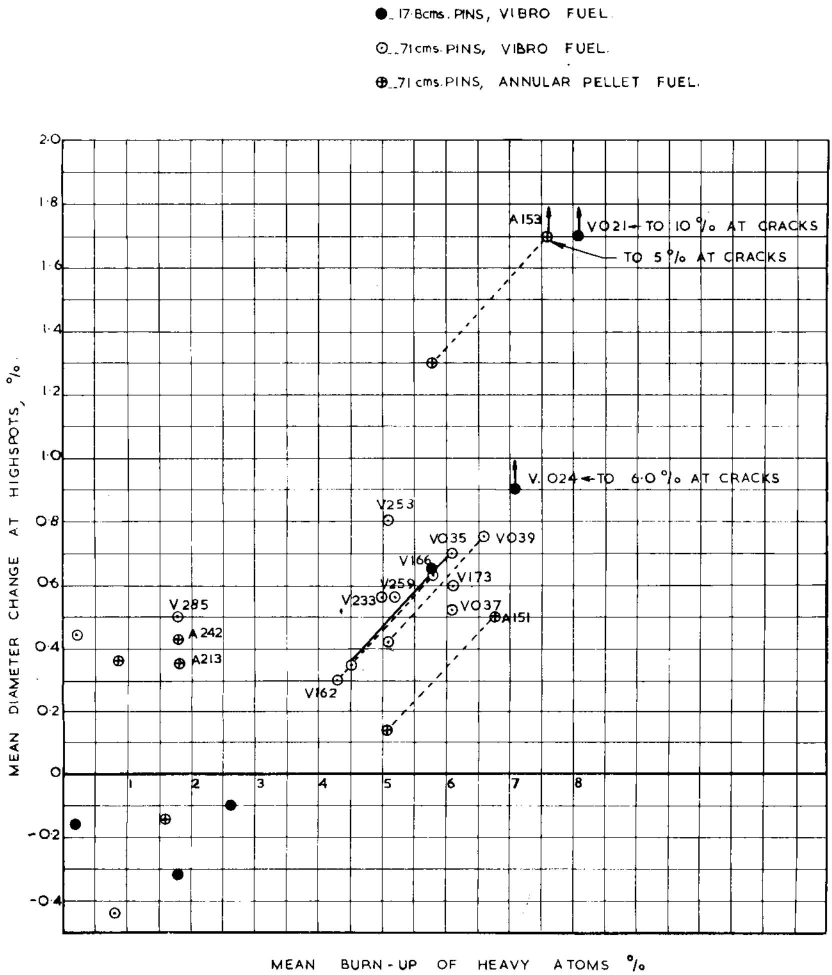


Fig. 5. Diameter changes in fuel pins irradiated in the DFR.

Fission Gas Release

Data on gas release and gas retention are quoted in Table 3 and gas release fraction is plotted against burnup in Fig. 6. The values quoted for pin V.035 in Table 3 are given as a range rather than single figures because they were

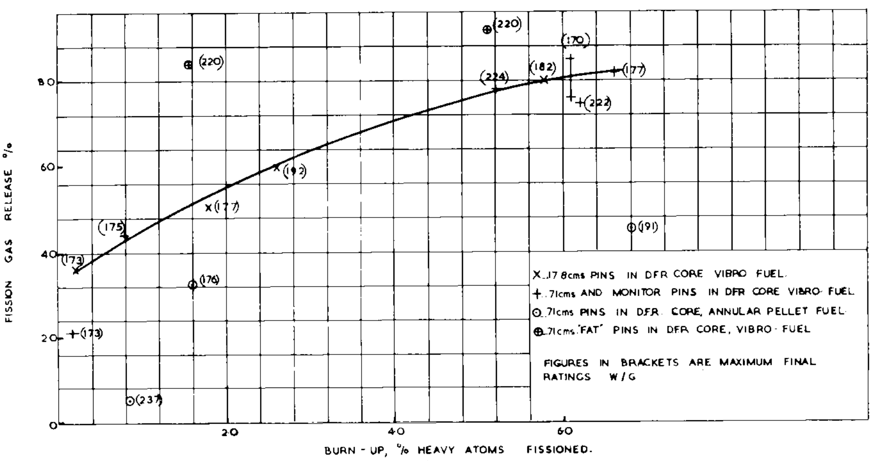


FIG. 6. Fission gas release data for mixed oxide (U 15% Pu)O_2, fuel in the DFR.

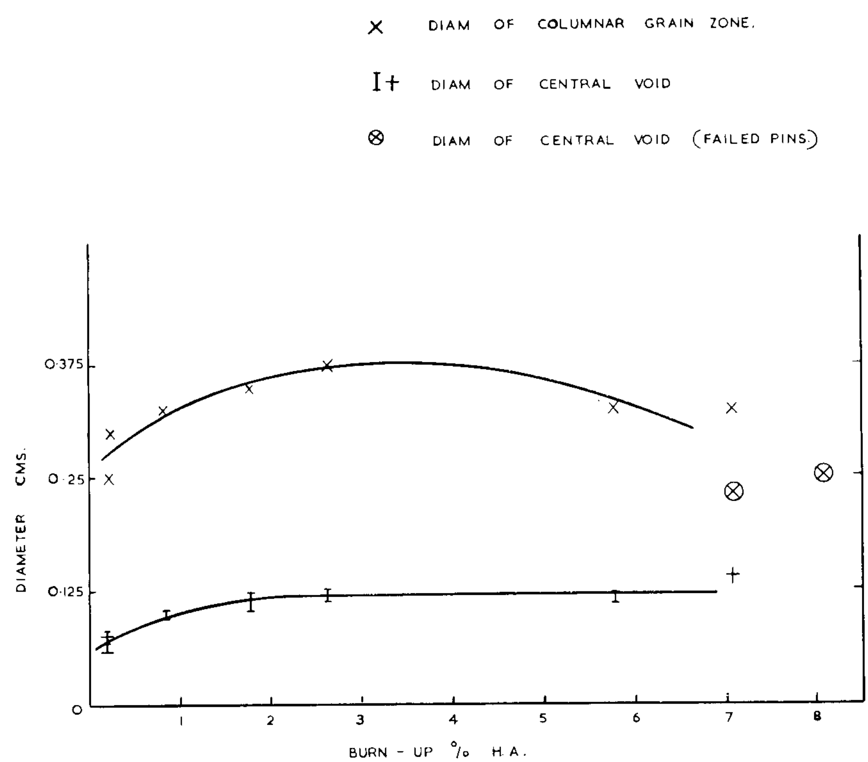


FIG. 7. Dependence of void and columnar grain zone diameters on fuel burnup.

calculated from the helium/fission gas ratio in the gas sample extracted after piercing the pin rather than from the quantity of fission gases collected. The discrepancy between the values estimated from the alternative methods, 80% compared with ~25%, indicates leakage of the pin at some late stage in irradiation or during examination, and for this reason this pin is accounted a possible failure.

It can be seen from Fig. 6 that the plot of gas release values from vibro pins against burnup follows a simple curve apparently insensitive to heat rating (above 170 W/g), cladding temperature or pin length. The plot is almost linear to exposures up to ~3%, and although it flattens at higher burnup, it seems probable that 85% to 90% gas release will be obtained at about 10% burnup. A gas release in excess of 90% is demonstrated by the 6·98 mm i.d. vibro pin at 5·2% burnup. Annular pellets release considerably less gas than the vibro-compacted variety, presumably because of lower fuel temperatures, and a maximum release of about 50% at 10% burnup appears probable. It must be expected therefore that in annular pellet oxide pins up to ~7·0 mm i.d. the principal can stressing mode will derive not from gas pressures in the plenum but from swelling of the fuel.

Consideration of the gas release values from vibro fuels in conjunction with metallographic data suggest that up to about 3% burnup the fission gas may be regarded as being released almost entirely from the columnar grain zone, a negligible amount escaping from the fuel outside this region. At higher burnup, this simple correlation with fuel structure breaks down, since although the gas release fraction continues to rise as burnup proceeds, the columnar grain zone diameter reaches a maximum value and then begins to contract as irradiation continues (Fig. 7).

Metallographic Examination

(*a*) *Vibro-compacted oxide*

The structure of vibro-compacted fuel showed progressive alteration as irradiation continued, but certain common features were distinguishable in all pins between 0·2% and 8%. These were:

(i) A central void, normally cylindrical in short pins, tapering gradually to small diameter at each end in long pins, with its course occasionally deviating from the pin axis.

(ii) An envelope of radially oriented ("columnar") grains surrounding the void, representing the extent of fuel densification by thermal sintering. Assuming that columnar grain growth occurs at temperatures above ~1700°C, the location of the columnar zone boundary suggests maximum fuel temperatures in the range 2000–2200°C.

(iii) An outer equi-axed grain zone, the grain size decreasing from the boundary of the columnar zone to the fuel/can interface; virtually no fuel densification occurs in this region, the porosity at high burnup corresponding reasonably well with that predicted from the initial fuel density.

Estimates of the void and columnar grain diameters as a function of burnup are plotted in Fig. 7. A system of circumferential, radial, and trans-

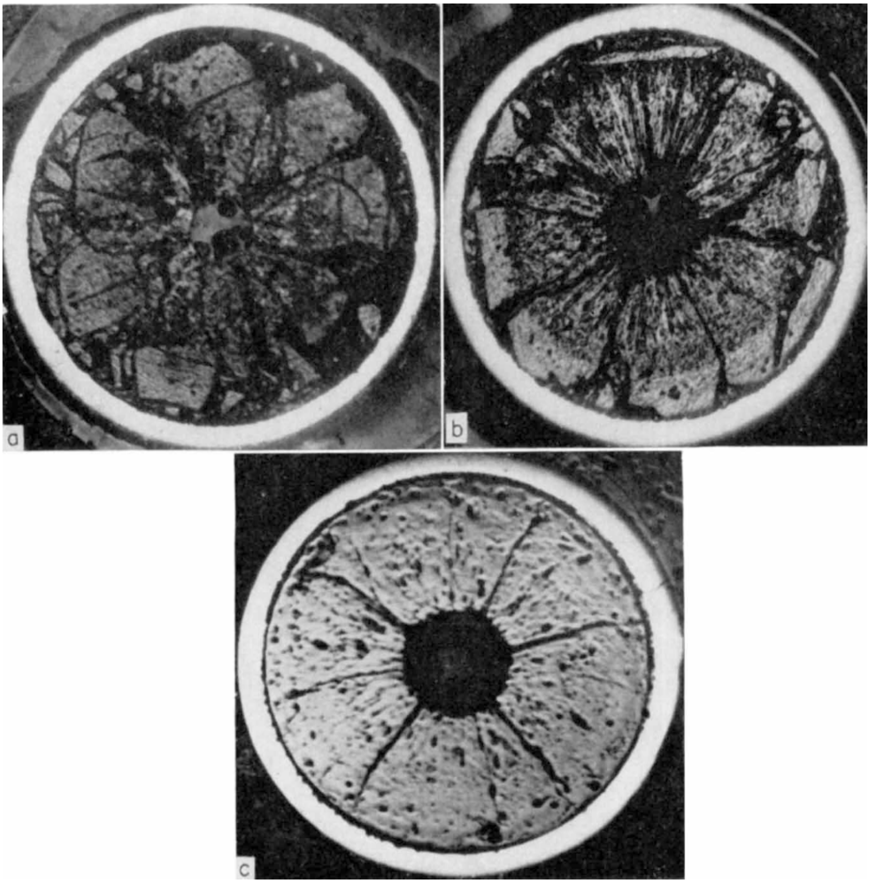


FIG. 8a. Transverse section through vibro fuel pin after 0·2% burnup. × 12.
FIG. 8b. Transverse section through vibro fuel pin, 2·6% burnup: as polished. × 12.
FIG. 8c. Transverse section through vibro fuel pin, 6·2% burnup. × 12.

verse cracks is also evident in all vibro pins, the first appearing to be a consequence of differential thermal expansion between the sintered core and a semi-coherent outer zone, the others probably resulting from shrinkage during sintering.

One notable difference between high and low burnup pins concerned the extent of fuel consolidation at the cladding interface. As shown in Figs. 8a–8c, the initial loose particle conglomerate coalesces into a coherent homogeneous mass as burnup progresses, even though the temperature of the fuel at the can surface could not have exceeded ~800°C. The fuel appeared to be pressed tightly against, and into, the clad, being so firmly bonded in some areas as to cause the surface layer to part from the fuel substrate rather than the can on cooling (Fig. 9). Such macroscopic structural changes are

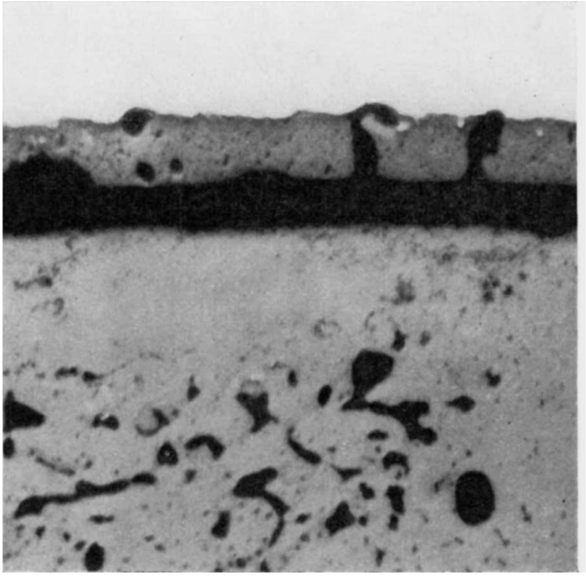


FIG. 9. Layer of fuel stuck to can. Pin V.162, 6·2% burnup. ×250.

associated with the nucleation and growth of porosity in the fuel annulus outside the columnar grain zone and a concept emerges of fuel swelling due to the growth of bubbles of fission gases which gradually absorbs inter-particle porosity and brings about pressure bonding of fuel/fuel and fuel/clad contact surfaces.

Small rounded inclusions, identified provisionally as fission product segregates (e.g. Ru, Mo) were abundantly present in the columnar grains and to a lesser extent outside this zone. Such inclusions were typically located at grain boundaries or pore surfaces and had a diameter of 10–15 μ at 6% burnup. A feature of the columnar grain zone seen only at high burnup was the development of sub-grain structures at the outer extremities of the elongated grains, a phenomenon which seems to constitute a stage in the regression of columnar grains to an equi-axed structure. Assuming that the boundary of the columnar grain zone represents a temperature isotherm, the variation in the diameter of this zone with burnup requires for its explanation an initial decrease in smear fuel conductivity followed by a gradual improvement at higher burnup. This sequence can be explained if it is assumed that the fuel/can temperature drop is a dominant factor determining the fuel temperature. The conductivity of this gas gap deteriorates as the pristine helium is contaminated with fission gases, and fuel temperatures therefore rise and the columnar grains elongate further. As fuel swelling proceeds, the effective width of the gas gap decreases, the fuel/can contact resistance is reduced, the fuel temperature therefore diminishes and in consequence the outer portions of the columnar grains cease to be rejuvenated by the passage of elliptical pores and revert to the equi-axed condition.

The behaviour of the natural UO_2 pellets was unexceptionable in all pins prior to V.021. Metallographic examination revealed that both UO_2 pellets in this pin were initially of low density, and that, uncharacteristically, the bottom pellet had sintered to form a reasonably solid annulus in close contact with the clad. It is surmised that a ratcheting mechanism involving the differential expansion between fuel and clad, plus a compulsive propensity for the fuel to stay in contact with the can at higher temperatures, is responsible for the multiple fracture at this point.

(*b*) *Annular pellet oxide*

No evidence of columnar grain growth has so far been seen in annular oxide fuel pins (of 5·08 mm i.d.) and temperature estimates suggest a maximum fuel temperature of $\sim$1600°C. Metallographic features of interest in fuel of this type were:

(i) The development of porosity as burnup increased, pores appearing first near the inner surface of the annulus, later at the outer surface. Porosity at high burnup tended to be fairly evenly distributed but was finer near the can except in the vicinity of cracks.

(ii) The gap between the fuel pellet and the can disappeared at high burnup; hence, as in the case of vibro fuel, fuel temperatures might be expected to increase and then decrease as the effects of fission gas contamination were gradually effaced by fuel swelling. Pellet–pellet gaps were still clearly distinguishable at $\sim$8% burnup.

(iii) The centre void diameter decreased over most of the fuel length of pin A.153, the reduction being least in the vicinity of the cracks. This inward distension is a clear demonstration that fuel swelling is occurring, and its magnitude suggests that the resistance to deformation of the cladding is, initially, considerably greater than that of the internal fuel surface.

As in the case of vibro fuel, a mechanism of fuel swelling based on volume changes resulting from solid fission product accumulation plus the growth of fission gas bubbles appears probable. A graphic illustration of the influence of can restraint on bubble growth is shown in Figs. 10a and 10b, relief of

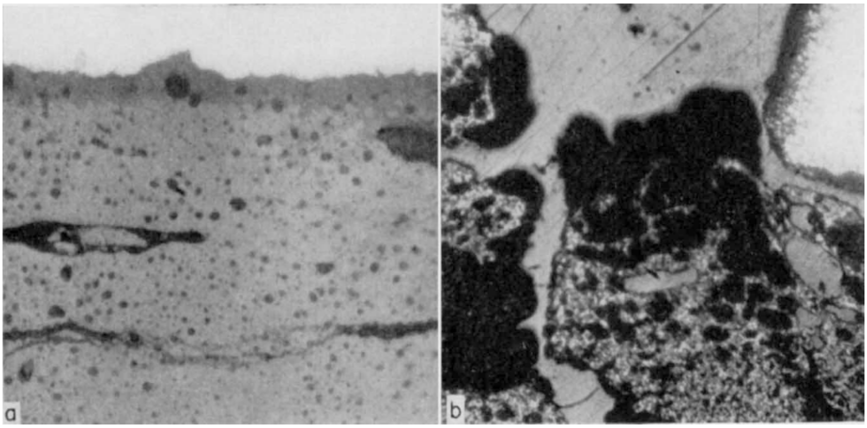


FIG. 10a. Fuel-cladding interface near middle of A.153. × 500.
FIG. 10b. Gross porosity near top crack in A.153. × 100.

FIG. 11. Inner surface of cladding V.021; etch, HF/HNO_3. ×250.

the constraint consequent on cracking of the can resulting in extraordinary enlargement of the pores near the fuel oxide surface. The distribution of inclusions resembled that in vibro oxides but segregates tended to be smaller, $\sim 6\ \mu$ diameter, and correspondingly more numerous.

(c) Cladding

Examination of the cladding on those oxide pins which were intact after irradiation showed few unexpected features and none which would lead to

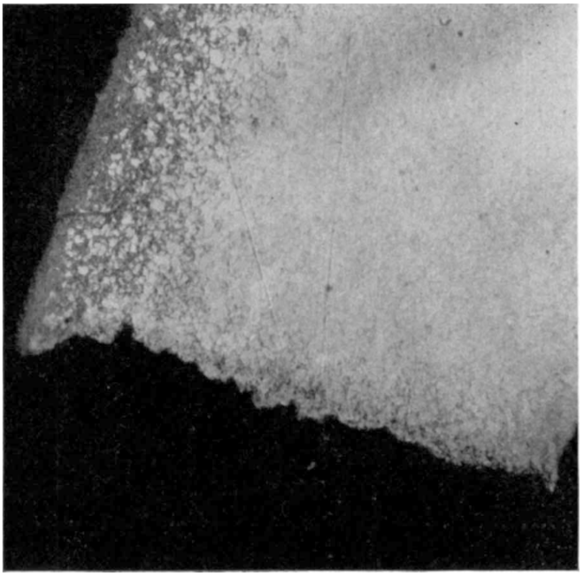


FIG. 12. Cladding at bottom crack in A.153; etch, picric acid/Hcl. $\times 200$.

premature failure of the clad. The cladding on failed pins showed several features relevant to the failure mechanism:

(i) Abnormally high clad temperatures were indicated in V.024 by the occurrence of grain growth plus sigma phase precipitation in the matrix, suggesting clad temperatures of $\sim 800°C$ rather than 700°C. Less certain evidence of temperatures higher than predicted was seen in the cladding of V.021 also.

(ii) Evidence of intergranular attack and crack formation on the inner surface of V.021 (Fig. 11).

(iii) Cracks appeared to follow an intergranular mode at the mid-points of the shorter pins, and showed a transgranular habit in the fractures of A.153 and at the base of V.021.

(iv) Some signs of carbon pick up from the coolant were seen at the base of A.153 (Fig. 12).

Failure Mechanism

In the early evaluations of ceramic fuel pin designs it was recognized that failure of the cladding might occur as a result of two principal straining mechanisms, i.e.:

(i) Strain induced by the pressure of fission gases escaping completely from the fuel, each gram of fuel generating about 0·210 cm^3 of gas at NTP per 1% burnup.
(ii) Strain caused by the swelling of the fuel induced by inexorable volume changes associated with the accumulation of solid fission products (accounting for volume increases of perhaps 1% per 1% burnup in oxides) plus the more controllable dilation due to the growth of fission gas bubbles trapped in the fuel matrix.

The effect of (i) can be minimized by provision of an adequate plenum but the magnitude and consequences of (ii) are not yet predictable, and the effect of such safeguards as the provision of voidage in the fuel could only be determined by experiment. The contributions of these two mechanisms to the three failures observed are not entirely clear, but the following points can be made:

(i) Multiple cracking is observed in both V.021 and A.153 and less certainly V.024; although gas pressure straining could initiate a number of cracks, loss of pressure when the first crack completes its traverse could effectively halt further propagation of the remainder.
(ii) The pins invariably show diameter increases very considerably in excess of those which would be predicted from pressure stresses in the pin unless:
   (a) cladding temperatures were $\sim 100°C$ higher than intended, or
   (b) creep of the cladding is vastly accelerated under irradiation, or
   (c) fission gas pressures are considerably increased, e.g. by isolation of portions of the fuel from the plenum, or
   (d) the cladding is markedly weakened by general or intergranular corrosion.

There is no evidence to support a contention that any of the last three possibilities had a large effect in magnifying the cladding strain rate. There is, however, fairly conclusive metallographic evidence that V.024 ran hot, and the observation that the plenum in this pin showed appreciable strain is strong inferential evidence for a gas pressure straining mechanism. Although the evidence for abnormally high temperatures in V.021 is less convincing, the similarity in failure mode (at the centre of the pin), and the fact that this pin too showed plenum expansion, argue in favour of a gas pressure straining mechanism being responsible for the failure of this pin also. In both cases, however, the fuel was intimately in contact with the clad at the end of irradiation, and the possibility that cracking was assisted or extended by fuel swelling cannot be ignored.

Failure of the cladding on pin A.153 can in no circumstances be attributed to gas pressure stressing. In the region of the upper and middle cracks, the clad temperatures were so low that creep strains would have been negligible. At the base of the pin also, strain should have been insignificant, since limiting internal pressures are fixed on the assumption that the can has the creep characteristics of solution treated material even when a much stronger cold-worked can is used. A fuel swelling mechanism must, therefore, be postulated to account for the failure. The mechanism proposed suggests that retention of fission gases in the fuel matrix—in which they are virtually insoluble—is achieved by the nucleation and growth of bubbles. At first these bubbles are of sub-microscopic size and therefore in equilibrium with surface tension forces, but as the concentration of fission gases increases, the bubbles grow and surface tension becomes decreasingly effective in opposing internal pressure stresses. After this stage, bubble enlargement is accompanied by visible distension of the fuel, at a rate determined mainly by the creep strength of the fuel matrix, until the fuel touches the clad. After contact is established, the direction of fuel growth will depend upon the resistance to deformation offered on one side by the clad, on the other by free surfaces in the fuel, e.g. between vibro-compacted particles or at the centre void.

When the fuel temperature is high, and the fuel interior reasonably plastic, deformation of the fuel will predominate, but as the stresses required to constrict a progressively smaller hole build up, the can itself is compelled to make provision for fuel swelling by its own expansion. The observation that the can is deformed even at low temperatures suggests swelling stresses approaching the yield stress of the clad, and it is feasible that the can yields progressively under slightly increasing stress—as swelling stresses increase—rather than creeps under a uniform stress. Assuming that the fuel swelling is largely dictated by retained fission gases, it follows that the location of the fracture will be determined by the interacting influences of such factors as burnup, can temperature, fuel temperature, and gas release fraction. Hence the cooler portion of a pin, in which the gas retention is high, may be more vulnerable than the higher burnup sections in which gas release is enhanced. Increase in the burnup to failure would, on the hypothesis outlined, be expected to result from an increase in cladding strength, a reduction in fuel density, promotion of higher gas release, and possibly by reducing the fuel strength so long as this alteration is not associated with an increase in gas-retaining propensities. The evidence of partial closure of the centre void in pin A.153 is encouraging in that it demonstrates that the fuel is plastic and that fuel strain may be directed inwards by a reasonably strong can. The association of fuel swelling with the growth of fission gas bubbles is also indirect evidence in support of the thesis that fuel swelling may be controlled by a strong clad. It must be recognized, however, that the use of fuels in which the gas retention is high may reduce the probability of failure by the fission gas pressure mechanism only at the risk of premature failure as a result of fuel swelling.

4. Conclusions

Evidence has been obtained that plutonium-bearing oxide fuel pins are capable of reaching mean burnups in excess of 5%, under PFR-type irradiation conditions. Above 7% mean burnup, fuel swelling causes pin diameters to increase, and this now appears to be the most serious problem to be overcome before 10% can be reached satisfactorily.

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